

Re: Response to Labor Market Openness Methodological Evaluation

Dear Professor Leblang,

Thank you for your thorough and thoughtful evaluation of the Labor Market Openness subindex. Your critiques are valuable, and we appreciate the care you took in reviewing this element of our Index. We have made substantive changes and additions to this subindex based on your recommendations and hope they sufficiently address your concerns. Where we made no changes, we provide a detailed explanation. Below are our responses to your feedback and recommendations.

I. Theoretical Framework

“There is an unresolved tension between “flexibility” (which often implies minimal regulation) and “protections” (which require regulation). The framework does not adequately explain how these potentially competing values are balanced or weighted in the overall construct.”

The tension between labor market flexibility and labor protections is correctly identified. Our intent by including them in one subindex is to more accurately capture the necessary balance between free deployment of human capital and its sustainability in the marketplace. Without protections from forced labor, coercion, or inequitable arrangements, providers of human capital are unable to deploy it as they see fit. Therefore, this subindex does not conceptualize market flexibility and protections as opposing poles, but rather as crucial institutional complements. Our adaptive PCA process is designed to accommodate the potential tension in these conceptual “buckets” by deriving principal components based on variation in the subindex’s entire collection of data. Our approach allows us to distinguish countries that fall into different cells of a 2×2 matrix: high flexibility/high protections (flexicurity models), high flexibility/low protections (minimal regulation models), low flexibility/high protections (rigid labor markets with strong unions), and low flexibility/low protections (dysfunctional markets). By deriving weights empirically through PCA rather than assuming these dimensions oppose each other, we can reward countries that achieve both dimensions well. The factor structure reveals whether these dimensions actually trade off in practice or can coexist.

“The framework omits several important dimensions of labor market openness: Occupational licensing and professional credential recognition, Non-compete agreements and other contractual restrictions on mobility, Monopsony power and labor market concentration, Informal sector dynamics, which are critical in many developing economies, Digital labor platforms and gig economy considerations”

The identified missing dimensions represent important areas for future development. We conducted extensive searches for cross-national data on these concepts:

- Occupational Licensing and Credential Recognition: Existing measures (e.g., OECD PMR licensing indicators) are limited primarily to OECD economies and lack country coverage. No global, annually updated dataset captures licensing restrictiveness in a comparable way across advanced and developing economies.
- Non-Compete Agreements and Contractual Mobility Restrictions: Data exist at the U.S. state level and in selected country case studies, but there is no cross-national index measuring enforceability or prevalence of non-compete clauses over time.

- **Monopsony Power and Labor Market Concentration:** While the economics literature has developed firm-level concentration measures, these rely on country-specific data and are almost exclusively deployed in advanced economies.
- **Informal Sector Dynamics:** The ILO publishes estimates of informal employment shares, but we hesitated to include them because their conceptual interpretation within the GCI is not straightforward. Our working intuition is that higher levels of informal employment indicate a larger share of workers operating outside formal contracts and institutionalized labor protections, which reflects weaker overall labor markets. However, informal sectors can also reflect structural features of developing economies rather than direct policy choices or institutional design. We would value any further suggestions on how best to embed informal employment into the GCI in a way that remains theoretically coherent.
- **Digital Labor Platforms and Gig Economy Regulation:** This is an emerging domain with no standardized global dataset capturing the impact of regulatory policy or labor platforms.

Because our framework prioritizes cross-national comparability, temporal consistency, and statistical reliability, we have chosen to include only indicators meeting those criteria. We view the dimensions identified as promising avenues for future refinement as global datasets improve.

“The application of creative destruction to labor markets may be theoretically questionable. While firm turnover drives innovation, rapid labor turnover can undermine firm-specific human capital investment and reduce productivity.”

The concern regarding the application of a Schumpeterian conceptualization of creative destruction to labor markets is also valid. We agree that excessive labor market turnover can be antithetical to well-functioning markets. For this reason, the indicator is based on survey metrics of turnover capacity and optionality, not the nominal level of labor churn. By measuring whether workers have the option to move when beneficial (true flexibility) rather than whether they are forced to move frequently due to instability (excessive churn), our indicators capture the positive aspects of labor market dynamism while avoiding the human capital erosion concerns you raise.

II. Component Validity

A. Labor Market Flexibility Component

“This component relies entirely on WEF Executive Opinion Survey data. While these measures capture important perceptions of labor market flexibility, they suffer from several limitations”

This component’s limitations, caused by relying on entirely on WEF survey data, are valid. Ideally, we would have global indicators like those suggested, including active labor market spending as a % of GDP, labor turnover rates, or other policy indicators. Unfortunately, these types of indicators do not exist for countries beyond those covered by the OECD metrics. These limitations will be clearly laid out in the GCI Whitepaper appendix.

B. Labor Protections Component

“Only two WEF indicators here, but they still introduce perceptual bias. Missing comprehensive measures of: Child labor prevalence, Workplace safety regulations and enforcement, Minimum wage adequacy, Union density and collective bargaining coverage,

Workplace discrimination beyond gender (race, ethnicity, disability, age)”

We believe these two WEF indicators add additional insight into the level of labor protection in a given market. Their relatively lower reliability scores reflect potential elite bias or survey consistency problems.

Regarding the missing comprehensive measures:

- **Child Labor and Forced Labor:** We reviewed the suggested Walk Free Global Slavery Index as a great addition, however, it is only published every 3-5 years, which disrupts the panel structure of our PCA analysis and would not add much beyond the ILO and V-DEM freedom from forced labor indicators.
- **Workplace Safety Regulations and Enforcement:** Comparable cross-national annual enforcement datasets are limited. We proxy enforcement capacity through the inclusion of the V-DEM Freedom From Forced Labor, which indirectly capture states’ willingness to enforce workplace standards.
- **Minimum Wage Adequacy:** Cross-national comparability remains challenging because statutory minimum wages vary in coverage, sectoral exemptions, and enforcement mechanisms. While nominal minimum wage levels are available, adequacy relative to median income or living wage benchmarks varies by methodology and is inconsistently reported across developing economies. We would welcome your thoughts on whether a standardized minimum wage-to-median wage ratio would sufficiently proxy global wage adequacy.
- **Union Density and Collective Bargaining Coverage:** We recognize the importance of organized labor in shaping labor protections. The ILO Compliance with Labor Rights dataset is based on their measurement of freedom of association and collective bargaining. However, union density remains difficult to measure globally. The only global dataset is the ILO’s Trade Union Density Rate, which, unfortunately, has not been updated beyond 2020.
- **Workplace Discrimination Beyond Gender:** We agree that broader anti-discrimination protections are important inclusions for this subindex. Unfortunately, globally comparable measures remain sparse outside high-income OECD contexts.

While we acknowledge that no single component or dataset fully captures the multidimensional nature of labor protections, we prioritize cross-national comparability, annual coverage, and conceptual alignment with the subindex. We will continue to refine this component as new datasets become available.

C. Labor Migration Component

“This component requires complete reconceptualization. If the goal is to measure labor migration openness, direct policy measures are essential. If such measures are unavailable, this component should be renamed to reflect what it actually measures (e.g., “Labor Market Outcomes” or “Perceived Labor Availability”).”

The disconnect between this component title and the actual measured concept is a valid critique. As a result, we re-titled this component to “Labor Availability” to better represent the concept. We agree that aggregate net migration is an outcome variable influenced by economic cycles, conflict, and humanitarian displacement, and therefore is not, on its own, a valid proxy for labor migration policy openness. Unfortunately, no measurements of migration policy exist at a global

level and this component must be proxied at some level. To account for the potential bias, we removed the net migration dataset in favor of ILO data on foreign-born workers as a share of total workers. The OECD Migration Policy Index is limited to OECD countries and datasets covering temporary work arrangements, visa policies, or bilateral labor agreements are difficult to translate to global metrics. Remittance flow data could potentially be useful for this component but runs the risk of over-weighting policy outcomes rather than inputs. We recognize that even with these improvements, this renamed component remains imperfect as it measures labor market availability outcomes rather than migration policy openness. However, given global data constraints, we believe 'Labor Availability' more accurately describes what we can actually measure while still providing useful information about labor market conditions.

III. Data Quality and Source Selection

The writeup of data quality and concerns in **Section IV Data Source Quality Assessment** of the review is a helpful summary. Our reliability scores are designed to account for these differences in data quality during the PCA process. In general, we will seek to add more ILO, V-Dem, and World Bank indicators as they are released. In particular, we are interested in adding the World Bank's new B-Ready survey which is gradually expanding to our full country list over the several annual rollouts.

IV. Methodological Approach

Before addressing your specific requests, we need to clarify an important terminological issue that may have caused confusion. In our documentation, we unfortunately used the term "components" to refer to the three conceptual groupings within the Labor Market Openness subindex: Labor Market Flexibility, Labor Protections, and Labor Migration. This was imprecise language on our part.

These three groupings are not statistical components in the technical sense. They are conceptual categories or "buckets" that organize our indicators thematically. The actual statistical components are derived empirically through PCA and do not necessarily align with these conceptual divisions. Here is how our methodology actually works:

When we apply PCA to the Labor Market Openness subindex, we input all 14 indicators simultaneously (the five WEF indicators you identified in Labor Market Flexibility, the four indicators in Labor Protections, and the four indicators in Labor Migration). The PCA process then extracts principal components based on the empirical covariance structure of these 14 variables. We typically extract two factors using oblimin rotation (oblimin being a recent alteration).

The resulting principal components reflect the underlying dimensions of variation in the data. These statistically-derived components may or may not correspond to our conceptual categories. For example, if certain "Labor Protections" indicators covary strongly with certain "Labor Market Flexibility" indicators, they may load on the same principal component, cutting across our conceptual boundaries.

This is one strength of the PCA approach. Rather than imposing our theoretical structure on the data (which would require arbitrary weighting), we allow the empirical relationships among indicators to reveal the actual dimensions of labor market openness. The conceptual categories

serve as an organizational framework for selecting indicators, but the statistical structure emerges from the data itself.

We apologize for the confusion our imprecise terminology may have caused. When you requested "within-component correlations" and "component-level standard errors," you were likely thinking of our three conceptual categories. In reality, our bootstrap procedures and correlation matrices operate on the actual principal components extracted from all 14 indicators jointly, not on these conceptual groupings separately.

With this clarification in mind, let me address your specific methodological requests:

A. Measurement Uncertainty and Standard Errors

You requested that we report standard errors or confidence intervals for all measures at multiple levels. This includes what you termed "component-level" and subindex-level standard errors, incorporation of source-provided uncertainty estimates, and temporal stability analysis.

Given the clarification above about our terminology, let me explain what we actually provide. Our methodology produces comprehensive uncertainty quantification that addresses your requests, though applied to the statistical principal components rather than conceptual categories.

Our methodology employs comprehensive bootstrap procedures specifically designed to quantify uncertainty at all levels of index construction:

- **Subindex Bootstrap Procedure:** For each subindex, including Labor Market Openness, we generate bootstrap confidence intervals through 1,000 to 2,000 iterations (depending on the structure of the subindex) with resampling. This procedure re-estimates the reliability-adjusted correlation matrix, applies PCA with oblimin rotation, and scores observations using bootstrap factor loadings.
- **Confidence Interval Calculation:** We calculate 95% confidence intervals using percentile methods from the bootstrap distribution. We provide the 2.5th and 97.5th percentiles for each country-year observation.
- **Source-Provided Uncertainty:** Where available (particularly for V-Dem data), we incorporate source-provided uncertainty estimates into the PCA process and propagate these through to final scores.
- **Outlier Handling:** We handle outlier bootstrap replicates through winsorization at extreme values to improve interval stability. When insufficient bootstrap samples are available, we construct conservative confidence intervals using normal approximations.

These bootstrap confidence intervals allow us to distinguish meaningful differences between countries and changes over time from measurement noise. By comparing whether year-to-year changes exceed bootstrap confidence intervals, we can determine whether observed changes represent meaningful policy shifts or measurement noise.

B. Correlation Matrices and Component Analysis

You requested comprehensive correlation matrices showing within-component correlations, cross-component correlations, and temporal stability of correlations. We provide the full 14×14 correlation matrix attached to our email, along with PCA loading matrices showing how each indicator

contributes to the extracted factors. Our email attachments also document the proportion of variance explained by each principal component and the factor loadings, allowing you to see exactly which indicators cluster together empirically and which contribute marginal versus central information.

C. Country Coverage Analysis

You requested detailed reporting on country coverage by year, indicator-specific coverage, missing data patterns, and coverage implications. We provide comprehensive coverage documentation:

- Annual country coverage (2008-2023) showing the number of countries with complete or missing data for each year
- Indicator-specific coverage showing the number of countries covered by each of the 14 indicators per year, disaggregated by geographic region and income level

Our adaptive PCA methodology directly addresses these coverage challenges:

- Period Identification: We compute the percentage of non-missing entries for each variable-year combination and group years with similar availability patterns using hierarchical clustering. This creates 2 distinct time periods, each with coherent data structures.
- Coverage Requirements: Variables are retained if they have 30% coverage within each period, with relaxation to 10% for specialized indicators. This adaptive threshold prevents both excessive data loss and inclusion of unreliable measures. Table X2 documents which indicators meet these thresholds in each period.
- Missing Data Robustness: We use pairwise correlation estimation to maximize data utilization while maintaining valid covariance structures. This allows us to include countries with partial data rather than requiring complete observations.

By clustering years with similar data availability and applying separate PCAs to each period, we avoid the spurious trends and selection biases that would arise from assuming constant coverage. This approach ensures that our scores reflect actual variation in labor market openness rather than artifacts of changing sample composition.

D. Temporal Variation Analysis

You requested documentation of how Labor Market Openness scores vary over time, including aggregate trends, country-specific trajectories, bucket-level trends, and temporal stability analysis. We provide this analysis in the email attachments.

- Aggregate trends showing the distribution of subindex scores across all countries for each year (2008-2023), including mean, median, standard deviation, minimum, maximum, and interquartile range
- Country-specific trajectories identifying countries with the largest positive changes, largest negative changes, highest stability, and highest volatility over the period
- Bucket-level trends tracking how each conceptual category (Labor Market Flexibility, Labor Protections, Labor Availability) evolves over time
- Time series plots showing temporal autocorrelation and year-over-year score changes with bootstrap confidence intervals

Your review expressed concern that year-specific normalization undermines temporal comparability. Our approach uses period-specific normalization based on data availability clustering, not arbitrary year-by-year normalization. Specifically:

- **Within-Period Standardization:** Factor scores are standardized within each temporal cluster, preserving relative rankings within time periods that have similar data structures.
- **Cross-Period Scaling:** Standardized scores are rescaled to a common 0-100 range using min-max normalization. This enables temporal comparisons while accounting for changing data availability across the 2 distinct periods we identified.
- **Outlier Treatment:** We winsorize extreme values at the 1st and 99th percentiles to reduce sensitivity to outliers that could distort temporal patterns.

This approach ensures that scores within periods with similar data structures are directly comparable, while still allowing meaningful cross-period comparisons through the harmonization process. The bootstrap confidence intervals allow us to distinguish meaningful policy changes from measurement artifacts. When year-to-year changes exceed the 95% confidence intervals, we can conclude they represent substantive shifts rather than statistical noise.

V. Recommendations for Improvement

Based on **Recommendations for Improvement** (section VI in the review) we considered adding ILO labor force participation data to the re-named Labor Availability component but decided to avoid adding more outcome metrics than needed. We removed the net migration dataset in favor of ILO data on foreign-born workers as a share of total workers using “Employed foreign citizens by sex and country of citizenship (thousands) – Annual.” Additionally, we considered adding ILO informal employment data to the Labor Protections component, wherein higher levels of informal employment can signal higher levels of vulnerability to harmful labor arrangements, but decided this dataset might be biased against developing labor markets.

Concluding Remarks

Thank you again for your careful review of the Labor Market Openness subindex. We apologize again for the confusion regarding the term “component”. However, we are happy to have had the opportunity to correct this error and provide you with the requested methodological documentation and underlying statistics. Below, we briefly summarize how your key concerns were addressed.

First, you raised a conceptual concern regarding the tension between labor market flexibility and labor protections. We clarified the theoretical framework to explicitly treat these dimensions as institutional complements and explained how our PCA methodology empirically determines whether these dimensions trade off or coexist across countries.

Second, you identified several important omitted dimensions of labor market openness. We conducted systematic searches for cross-national datasets in these areas. Where no globally comparable, annually updated data exist, we documented these constraints explicitly and identified them as priorities for future development rather than introducing weak proxies.

Third, you raised concerns about the Labor Migration component’s conceptual validity. In response, we reconceptualized and renamed this component as *Labor Availability*, removed net migration as

an outcome variable, and substituted ILO data on foreign-born workers to better align the indicator with what can be measured globally.

Fourth, you noted potential weaknesses related to data quality, reliance on WEF indicators, and perceptual bias in survey metrics. We clarified how reliability scores downweight these indicators in the PCA process, expanded our discussion of data limitations, and reaffirmed our preference for transparent treatment of imperfect data over inconsistent global coverage.

Finally, you requested additional methodological transparency regarding correlations, uncertainty, coverage, and temporal stability. We provided all requested documentation in the attached files.

We appreciate your engagement and believe these revisions significantly improve the Labor Market Openness subindex. We would be happy to discuss any of these points further and hope to continue this productive dialogue.

Sincerely,

UVA Democracy and Capitalism Lab